

By the End of Today You Will...

- Explain how a photo is stored as a grid of numbers
- Find an edge in a number grid with your eyes *and* with math
- Write and run a Python edge-detection filter
- Name one real-world place where this code is used right now

Vocabulary – 4 Key Words

Word	What it means
Pixel	The tiniest dot in a digital picture – each one stores a single number. Zoom in on any photo and you'll see the squares!
Edge	A place in the image where numbers change a lot very quickly – like the outline of a shape where light meets dark
Filter	A math operation that slides over every pixel and checks its neighbors – different filters blur, sharpen, or find edges
Nested Loop	A loop <i>inside</i> another loop – the outer loop moves through rows, the inner loop through columns, visiting every pixel one by one

Warm-Up (Write one sentence for each – no wrong answers!)

1. If you zoom in really close on a phone photo, what do you see? What color are those tiny squares?
2. Look at a photo of a cat on a white background. Where does the cat *end* and the background *begin*? What do we call that line?

Step 1 – Pictures = Numbers

A digital photo is just a **grid of numbers**. Each number = one pixel's brightness (0 = black, 9 = very bright).

text

Our 5x5 mini-picture:

```
[ 0, 0, 0, 0, 0 ]   ← dark border
[ 0, 5, 5, 5, 0 ]
[ 0, 5, 9, 5, 0 ]   ← 9 is the brightest pixel
[ 0, 5, 5, 5, 0 ]
[ 0, 0, 0, 0, 0 ]   ← dark border
```

The blue numbers are the bright square. The zeros are the dark border. **Can you see the square shape?**

Step 2 – What Is an Edge?

In a photo, an edge is where **dark meets bright** – like where a dark sky meets a bright building. Your eye sees a line there.

In numbers, going across a row:

text

0 → 0 → 5 → 5

The jump from **0 to 5** is an edge – a big number change!

In our 5x5 grid, find where numbers jump from 0 to 5. Point to it – that's where an edge lives!

Why AI needs edges: Edges define shapes → shapes tell AI "this is a face," "this is a stop sign," "this is a cat!"

Step 3 – The Edge Formula

For every pixel, ask: **"Are you very different from your 4 neighbors?"**

value=center×4-(up+down+left+right)

- **Small value** (close to 0) → center looks like its neighbors → flat zone, no edge
- **Big value** → center is very different → **EDGE!**

Let's calculate pixel together:

Variable	Value
center = image[1][1]	5
up = image[0][1]	0
down = image[2][1]	5
left = image[1][0]	0
right = image[1][2]	5

value = 5×4 - (0+5+0+5) = 10 → EDGE!

Now you try – pixel in your notebook:

9×4 = _ and 5+5+5+5 = _ so the answer = _

The Code – Type Along

python

```

# Our tiny picture (5 rows, 5 columns)
image = [
    [0, 0, 0, 0, 0],
    [0, 5, 5, 5, 0],
    [0, 5, 9, 5, 0],
    [0, 5, 5, 5, 0],
    [0, 0, 0, 0, 0],
]

filtered = [] # our results go here

for y in range(1, 4): # rows 1, 2, 3
    new_row = []
    for x in range(1, 4): # columns 1, 2, 3
        center = image[y][x]
        up = image[y-1][x]
        down = image[y+1][x]
        left = image[y][x-1]
        right = image[y][x+1]

        value = center*4 - (up + down + left + right)
        new_row.append(value)
    filtered.append(new_row)

```

print(filtered)

What each part does:

Line	What it means
Line	What it means
for y in range(1, 4)	Visit rows 1, 2, 3 – skip row 0 because y-1 would go off the grid!
for x in range(1, 4)	Visit columns 1, 2, 3 – same reason
image[y-1][x]	The pixel <i>above</i> the current one
center*4 - (...)	The edge formula – big result = big edge signal

Exercise A – Copy & Explore

1. Copy the code above into your Python editor (IDLE or Replit) and run it
2. How many rows and columns does filtered have?
3. Change image[2][2] = 9 to image[2][2] = 1 and run again – what changed and why?

Bonus: Can you change the image to make **all** filtered values equal to 0? (*Hint: what if every number in the grid is the same?*)

Exercise B – Fill in the Blanks

```

python
image = [
    [0, 0, 0, 0, 0],
    [0, 7, 7, 7, 0],
    [0, 7, 7, 7, 0],
    [0, 7, 7, 7, 0],
    [0, 0, 0, 0, 0]
]

```

Homework

Question 1 – Draw & Calculate:

Design your own 5x5 number grid with at least one clear edge (use 0 for dark, 5-9 for bright). Pick one inner pixel and calculate its filtered value by hand using the formula:

$value = center \times 4 - (up + down + left + right)$

Is it an edge pixel or a flat zone? How do you know?

Question 2 – Reflect in 3 Sentences:

Why do you think edge detection is useful for a computer trying to recognize a face or read a street sign? In your answer: (1) explain what an edge *is* in your own words, (2) give one specific example, and (3) look back at your warm-up answers – did today's lesson change how you think about photos?

Email a photo of your grid + your 3 sentences to **hue.weng.88@gmail.com** by next class.

Today's Lesson

<https://hue881.github.io/1/index.html>
<https://hue881.github.io/1/activity/index.html>
<https://github.com/hue881/AI-Coding-Day16/tree/main/1>